

TIC 258204238

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.687036 d
Epoch (t0) 1683.7856 BTJD
Duration 0.3500 d
Depth 1.132 % (11,319 ppm)
SNR 692.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

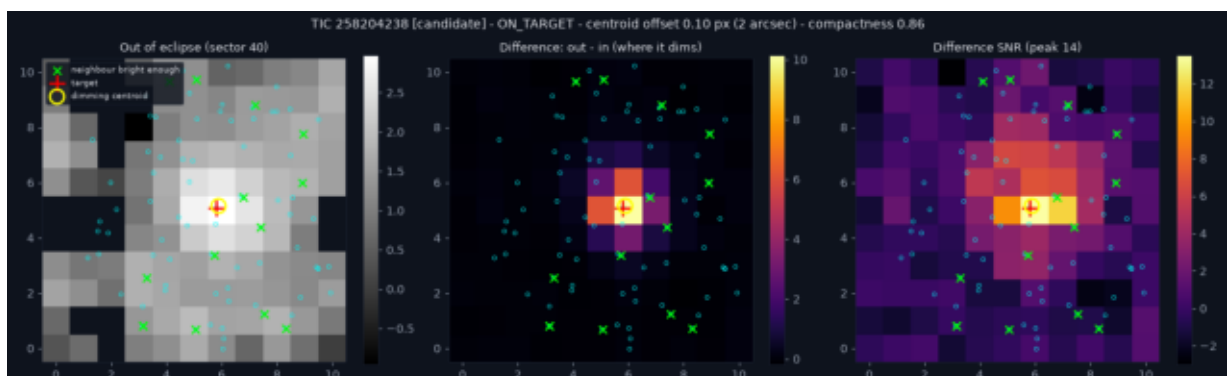
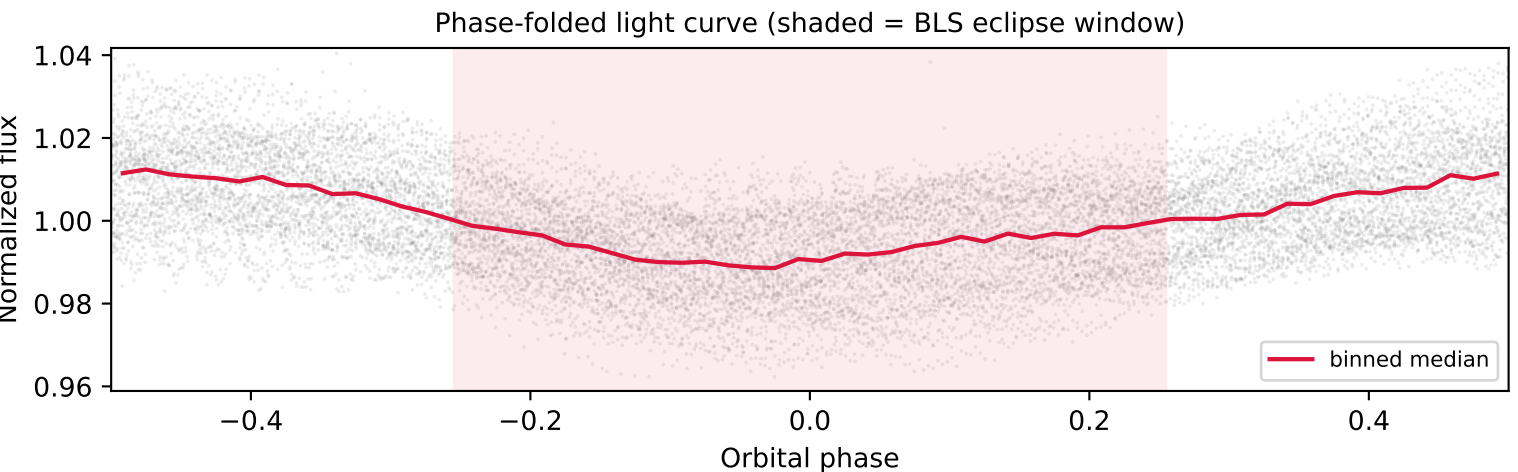
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 226 (gate ≥ 3)
Robust depth ratio 1.08 (gate ≥ 0.50)
Robust depth sigma 74.1 (gate ≥ 7.0)
Odd/even z-score 0.01 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 40 light-curve dip in this sector: 23.5 sigma t0 refined by 0.0475 phase
Difference-image peak SNR 13.50 (gate ≥ 5.0) Compactness 0.86 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 258204236
0.94 px away, dTmag +3.97 - farther from the dimming than the target



No caveats - clean detection on every check.